PREDICTIVE MAINTENANCE

AI/ML & AWS Cloud Project

PROJECT PLANNING & MANAGEMENT
LITERATURE REVIEW
REQUIREMENTS GATHERING

Prepared for submission: 16 October 2026

This document consolidates the three deliverables due on 16 October 2026, using the existing Predictive Maintenance proposal as the project baseline. It defines the plan, reviews relevant literature, and formalizes stakeholder and system requirements.

Project team

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Project assumption: the minimum viable product (MVP) will use a public or simulated machine-condition dataset. Real sensor integration through AWS IoT Core is an optional extension, not a prerequisite for the first working version.

Document Control

Field	Description
Project	Predictive Maintenance — AI/ML & AWS Cloud
Submission date	16 October 2026
Document scope	Project Planning & Management; Literature Review; Requirements Gathering
Baseline architecture	Dataset/simulator → Amazon S3 → Amazon SageMaker → SageMaker endpoint → AWS Lambda → API Gateway → web dashboard; CloudWatch monitoring; IAM security.
Status	Planning and requirements document. Model performance and deployment KPIs are targets to be verified, not claimed results.

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Important note: The provided course guideline does not include an official marking rubric or supervisor feedback. Those items are therefore represented as a feedback record to complete and a clearly labelled proposed internal rubric; they are not presented as official instructor decisions.

1. Project Planning & Management

1.1 Project Overview

Predictive Maintenance is a cloud-based AI/ML system that analyzes machine-condition data to estimate failure risk and identify abnormal operating behavior before a breakdown occurs. The intended users are maintenance personnel who need a data-supported way to prioritize inspections. The system supports human decisions; it does not autonomously authorize repairs or guarantee that failures will be prevented.

1.2 Problem Statement

Reactive maintenance may lead to unexpected downtime, emergency repair, production interruption, and secondary damage. Fixed-schedule preventive maintenance can replace healthy components or fail to detect a problem that develops between scheduled checks. The proposed system uses condition data and machine-learning models to flag risk earlier and help maintenance staff decide when inspection may be appropriate.

1.3 Objectives

• Prepare a documented dataset and reproducible preprocessing/feature-engineering pipeline.

• Train and compare a supervised baseline (Random Forest or Gradient Boosting/XGBoost) and, when useful, an anomaly detector such as Isolation Forest.

• Evaluate model behavior with Precision, Recall, F1-score, ROC-AUC, a confusion matrix, and false-alarm analysis.

• Store data and model artifacts in Amazon S3 and use Amazon SageMaker for training, evaluation, and inference hosting.

• Expose predictions through AWS Lambda and Amazon API Gateway and present results in a simple web dashboard.

• Use Amazon CloudWatch for logs and operational monitoring, and AWS IAM for least-privilege access.

• Deliver technical documentation, test evidence, deployment instructions, and a final demonstration.

1.4 Scope

In scope for the MVP	Out of scope / optional extension
Public or simulated data; data validation and preprocessing; feature engineering; model training and evaluation; S3 storage; SageMaker model/endpoint; Lambda + API Gateway prediction flow; dashboard; CloudWatch logs/metrics; IAM permissions; functional and integration testing.	Physical sensor procurement and installation; guaranteed industrial-grade failure prevention; 24/7 production service-level agreement; automatic maintenance scheduling; multi-site enterprise integration; RUL estimation unless dataset and time permit; IoT Core real-time ingestion if it delays the core MVP.

1.5 Assumptions and Constraints

• The initial dataset will be selected from a public source such as AI4I 2020 or NASA C-MAPSS, depending on whether the chosen task is failure classification or remaining-useful-life estimation.

• If failure labels are insufficient, anomaly detection may be used as a complementary approach; its scores must not be described as calibrated failure probabilities without validation.

• The team has limited student-project time and must keep AWS usage small, stop unused endpoints, and check current pricing before deployment.

• The system is a decision-support prototype. Maintenance personnel remain responsible for interpreting alerts and deciding on action.

2. Project Plan: Schedule, Gantt Chart, Milestones & Deliverables

The following schedule is aligned to the course guideline dates and the eight-week implementation outline in the original proposal. Dates after 16 October are planning targets and should be adjusted if the instructor changes the project calendar.

Phase / Workstream	Planned dates	Milestone / output	Lead
1. Requirements and dataset selection	16–20 Oct 2026	Requirements baseline; dataset decision; data dictionary outline	Sama + all
2. EDA, cleaning and feature engineering	21–27 Oct 2026	Clean dataset; preprocessing notebook/script; leakage-safe split	Sama + Abdelrahman Ahmed
3. Model development and evaluation	28 Oct–3 Nov 2026	Baseline and candidate models; metrics and error analysis	Abdelrahman Ahmed
4. AWS foundation	28 Oct–6 Nov 2026	S3 structure, IAM roles/policies, SageMaker setup, CloudWatch plan	Abdelrahman Sayed
5. Model deployment	7–13 Nov 2026	SageMaker endpoint and tested inference payload	Abdelrahman Sayed + Ahmed
6. API and dashboard integration	14–20 Nov 2026	Lambda/API Gateway flow; dashboard connected to API	Farida + Mariam Mohamed
7. End-to-end testing and monitoring	21–26 Nov 2026	Test report; failure-path checks; CloudWatch evidence	Mariam Ashraf + all
8. Documentation and final presentation	27 Nov–4 Dec 2026	Final report, deployment guide, demo and presentation	All

2.1 Gantt Chart (planning view)

Workstream	16–20 Oct	21–27 Oct	28 Oct–3 Nov	4–10 Nov	11–17 Nov	18–24 Nov	25 Nov–4 Dec
Requirements & dataset	●						
EDA / preprocessing	●	●					
ML experiments			●	●			
AWS foundation			●	●			
Deployment				●	●		
API / dashboard					●	●	
Testing / monitoring						●	●
Documentation / demo							●

2.2 Milestones and Acceptance Evidence

ID	Milestone	Evidence required
M1	Requirements baseline approved	Stakeholder list, use cases, FR/NFR catalogue, scope and acceptance criteria.
M2	Dataset ready	Source citation, data dictionary, quality checks, split strategy and preprocessing output.
M3	Model evaluated	Reproducible experiment notes, confusion matrix, Precision/Recall/F1/ROC-AUC where applicable, false-alarm analysis.
M4	Cloud inference works	SageMaker endpoint test, valid/invalid input tests, latency observations and logs.
M5	Application integrated	API request/response evidence and dashboard screenshot or demonstration.
M6	Final MVP accepted	End-to-end test report, risk closure notes, deployment instructions, final presentation.

3. Resource Allocation & Task Assignment

3.1 Team Roles

Team member	Primary responsibility	Key outputs
Abdelrahman Ahmed	AI/ML pipeline, feature engineering, model development and evaluation; ML integration.	Training/evaluation scripts, experiment log, model artifact, metrics report.
Abdelrahman Sayed	AWS infrastructure: S3, IAM, SageMaker environment and cloud deployment.	AWS setup, permissions, deployment configuration, cost controls.
Sama Samir	Data collection, data cleaning, EDA, preprocessing and dataset documentation.	Dataset source record, data dictionary, quality report, cleaned data.
Farida Montasser	Backend/API integration: Lambda, API Gateway and SageMaker communication.	Validated API flow, request/response handling, error paths.
Mariam Mohamed	Frontend/dashboard and machine-health visualization.	Dashboard views for sensor values, risk level, anomaly status and action guidance.
Mariam Ashraf	Testing, system integration, documentation, experiment organization and presentation support.	Test cases/results, integration checklist, document consistency and demo support.

3.2 Shared Responsibilities

• All members collaborate on Git/GitHub, code review, integration testing, progress reporting, and preparation of the final demonstration.

• Each workstream owner must document assumptions, blockers, completion evidence, and dependencies in the shared project tracker.

• Changes to the dataset, model target, AWS services, or project scope must be discussed by the team before implementation.

• No AWS credentials, access keys, or secrets may be committed to the repository. Use IAM roles and least-privilege permissions.

3.3 Resource Allocation

Resource	Allocation / use	Control
People	Six students assigned by primary workstream; cross-review at integration points.	Weekly progress review; backup reviewer for critical components.
Data	Public dataset or simulated telemetry; keep a versioned raw copy and processed outputs.	Record source/license, schema, label meaning, missing values and split strategy.
Software	Python, Pandas, NumPy, scikit-learn; optional XGBoost; Git/GitHub; HTML/CSS/JavaScript or Angular.	Pin dependencies where practical; keep setup instructions.
AWS	S3, SageMaker, Lambda, API Gateway, CloudWatch and IAM; IoT Core optional.	Budget alert if available, minimal instance sizes, shut down unused endpoint.
Time	Eight-week plan through final presentation on 4 Dec 2026.	Prioritize working end-to-end MVP over optional extensions.

4. Risk Assessment & Mitigation Plan

Risk ratings below are qualitative planning estimates, not measured probabilities. The team should review them weekly and update status as implementation evidence becomes available.

Risk	Likelihood / impact	Mitigation and contingency	Owner / trigger
Insufficient failure examples or poor-quality data	Medium / High	Validate schema, ranges, missing values, duplicates and labels; document data limitations; compare anomaly detection if labels are scarce. If data quality is not acceptable, select a better-supported public dataset or narrow the task.	Sama + Ahmed; trigger: too few positive failure cases or major data-quality defects.
Class imbalance causes missed failures	High / High	Use stratified or time-aware splits as appropriate; evaluate Recall, Precision, F1, confusion matrix and PR behavior; try class weights or resampling only within training data. Do not rely on accuracy alone.	Ahmed; trigger: poor minority-class recall.
Time-series leakage / optimistic test metrics	Medium / High	Split chronologically or by machine/time group where appropriate; fit preprocessing only on training data; keep test data untouched until final evaluation.	Ahmed + Sama; trigger: unusually high validation/test performance or shared time windows.
AWS connectivity, endpoint or service failure	Medium / High	Validate requests, set timeouts, handle exceptions and non-success responses, log correlation/request identifiers, and monitor errors/latency with CloudWatch. For demos, retain a local test path or saved sample predictions if cloud services are temporarily unavailable.	Farida + Sayed; trigger: endpoint/API errors or failed requests.
IAM permission errors or excessive permissions	Medium / High	Use least-privilege IAM roles; verify trust relationships and resource ARNs; test access with a minimal workflow; never hard-code credentials. Keep a documented permission checklist.	Sayed; trigger: AccessDenied or unexpected access.
AWS costs exceed student budget	Medium / High	Use small resources, train only when needed, stop/delete unused endpoints, avoid always-on GPU resources, review Cost Explorer/Pricing Calculator and set budget alerts where available.	Sayed + all; trigger: unexpected spend or idle resources.
Overengineering / missed deadline	Medium / High	Freeze the MVP scope; make IoT Core and QuickSight optional; deliver the S3-to-model-to-API-to-dashboard flow first. Defer advanced RUL or multi-site support.	Project team; trigger: core pipeline milestones slip.
False alarms reduce user trust	Medium / Medium	Tune thresholds using validation data; report false positives and false negatives; show uncertainty/limitations and require human inspection decisions.	Ahmed + Mariam Mohamed; trigger: excessive false alarms during testing.
Integration or schedule dependency delays	Medium / Medium	Define request/response schema early, use mock responses until endpoint is ready, integrate incrementally, and keep owners/due dates visible in the tracker.	All; trigger: blocked integration task >2 working days.
Security or accidental data/secret exposure	Low–Medium / High	Restrict S3 access, apply IAM least privilege, keep secrets out of Git, use environment/configuration management, and review repository before submission.	Sayed + all; trigger: exposed credential or unauthorized data access.

Risk response rule: if a high-impact risk threatens the MVP, prioritize a working and documented fallback over an untested optional feature. All exceptions and compromises should be recorded in the final report.

5. Key Performance Indicators (KPIs)

The following are proposed acceptance targets for a student MVP. They are not achieved results. Targets must be confirmed after the dataset and deployment environment are selected; all actual values should be filled in from experiments and test logs.

KPI	Proposed target / evaluation rule	Measurement method	Evidence / owner
Failure detection recall	Target ≥ 0.80 on a held-out test set, if labels and dataset support a meaningful estimate.	Recall = TP / (TP + FN); report confusion matrix and positive-class support.	Evaluation report; Abdelrahman Ahmed.
F1-score	Target ≥ 0.75 on the held-out test set as an initial project goal; revise only with documented justification.	F1 is the harmonic mean of Precision and Recall; compare candidate models consistently.	Experiment log; Abdelrahman Ahmed.
ROC-AUC	Report held-out ROC-AUC for binary classification when both classes are represented; compare against a baseline rather than assuming a guaranteed value.	Compute from continuous model scores/probabilities; note class and split details.	Evaluation report; Abdelrahman Ahmed.
API response latency	Proposed target: p95 ≤ 3 seconds for the deployed prediction API under the agreed demo test conditions.	Send a documented batch of representative requests; calculate p50/p95 and record region, endpoint status and sample count.	API test log; Farida + Sayed.
API request success	Proposed target: ≥ 95% successful valid requests during a controlled integration test; invalid requests should return a controlled error.	Count successful valid requests / total valid requests; separately verify invalid payload handling.	Integration test report; Farida.
System availability	Measure availability during scheduled demo/test windows; establish a target only after the test window and monitoring method are agreed.	CloudWatch metrics/logs and endpoint/API checks; exclude planned shutdowns only if documented.	Monitoring evidence; Sayed + Mariam Ashraf.
Prediction reliability	Report false-positive rate and false-negative count; no single threshold is accepted without reviewing the trade-off with maintenance users.	Confusion matrix and threshold comparison on validation data; final threshold chosen before test evaluation.	Model report; Ahmed + maintenance-user proxy.
Data processing reliability	100% of defined validation test cases should produce either a valid processed record or a clear, controlled rejection.	Run schema, missing-value, range, malformed-input and duplicate checks.	Data validation test results; Sama.
AWS cost control	No intentionally idle SageMaker endpoint left running after a work session; review actual usage before and after demos.	Resource checklist and billing/usage review; use alerts where available.	AWS checklist; Abdelrahman Sayed.

KPI reporting template

Metric	Target	Actual result	Pass / follow-up
Recall	≥ 0.80	To be measured	Complete after model evaluation
F1-score	≥ 0.75	To be measured	Complete after model evaluation
API p95 latency	≤ 3 seconds	To be measured	Complete after deployment test
Valid-request success	≥ 95%	To be measured	Complete after integration test

6. Literature Review

6.1 Purpose and Review Focus

This review summarizes research and guidance relevant to the proposed student MVP: machine-learning-based predictive maintenance, dataset selection, evaluation under rare failures, and deployment monitoring. It is a focused narrative review rather than a claim to be an exhaustive systematic review. The team should record the final dataset and any additional papers consulted during implementation.

6.2 Summary of Relevant Literature

Source	Relevant contribution	Implication for this project
Dalzochio et al. (2020), Machine learning and reasoning for predictive maintenance in Industry 4.0	Reviews ML and reasoning approaches in predictive maintenance and highlights practical challenges including scalability, latency and data security.	The project should evaluate not only model quality but also API latency, AWS integration and security.
Meitz et al. (2025), A Literature Review Framework and Open Research Challenges for Predictive Maintenance in Industry 4.0	Reviews 249 publications and structures the field around data, sensor analysis, ML complexity and research challenges.	Supports a staged MVP: first validate data and task fit, then add model complexity only when justified.
NASA C-MAPSS research / NTRS review (2015)	Describes benchmark turbofan prognostics datasets and issues such as sensor noise, operating conditions and multiple fault modes.	C-MAPSS is more suitable if the team chooses remaining-useful-life/prognostics; it may not directly match a simple binary failure classifier without target definition.
Challenges of ML-based RUL prognosis on NASA C-MAPSS (IEEE ETFA, 2021)	Reviews RUL approaches and identifies interpretability, uncertainty and domain adaptation as important challenges.	If RUL is attempted, report task-specific metrics and limitations; keep RUL optional for the MVP.
Dadfarnia & Sharp (2022), NIST/IEEE work on risk-based evaluation of AI-driven condition monitoring	Argues that the application area, risk management process and monitoring mechanism matter when evaluating condition-monitoring impact.	Connect model metrics to maintenance decisions and monitoring, not just a single accuracy number.
Dadfarnia, Sharp & Herrmann (2025), systematic review of condition-monitoring evaluation	Reviews how condition-monitoring technologies are evaluated and notes the importance of assessing engineering and financial benefits.	Describe the system as decision support; avoid claiming avoided downtime or financial savings without real operational evidence.
NIST AI RMF and deployed-AI monitoring guidance	Provides risk-management framing and highlights post-deployment monitoring for functionality, operations and unexpected behavior.	Use explicit risks, measurable acceptance criteria, CloudWatch logs/metrics, and a process for reviewing errors after deployment.

6.3 Comparison of Candidate Approaches

Approach	Strengths	Limitations / fit	Decision for MVP
Random Forest	Strong tabular baseline; handles nonlinear relationships; relatively interpretable feature importance.	Can be biased by class imbalance; probability calibration and threshold still need evaluation.	Use as a first supervised baseline if labeled failure data is available.
Gradient Boosting / XGBoost	Often competitive on structured tabular data; captures complex feature interactions.	More tuning; can overfit if validation is not disciplined; extra dependency may be needed.	Compare against Random Forest; retain only if results justify complexity.
Isolation Forest	Can identify unusual observations when failure labels are scarce.	Anomaly score is not automatically a failure probability; anomalies may be harmless.	Use as fallback/complement when labels are insufficient; evaluate with available ground truth and expert review.
Remaining Useful Life (RUL)	Provides an estimate of time until maintenance/failure when dataset supports sequential degradation targets.	More complex target, data preparation and metrics; not interchangeable with binary classification.	Optional extension only after MVP works.

7. Feedback & Evaluation Record

The supplied proposal and guideline do not contain actual instructor comments or a completed supervisor evaluation. To avoid inventing feedback, use the record below during the next supervisor meeting and update it with the exact comments received.

Date	Reviewer	Feedback / issue raised	Action agreed	Status / evidence
To be completed	Course instructor / supervisor	Record the actual feedback here.	Assign an owner and due date for each action.	Open
To be completed	Course instructor / supervisor	Record the actual feedback here.	Assign an owner and due date for each action.	Open
To be completed	Course instructor / supervisor	Record the actual feedback here.	Assign an owner and due date for each action.	Open

7.1 Project Evaluation Plan

• Technical evaluation: compare candidate models using the same data split and report Precision, Recall, F1-score, ROC-AUC where appropriate, confusion matrix, and false-alarm behavior.

• Data evaluation: document source, target definition, class balance, missingness, duplicate records, leakage risks, and preprocessing decisions.

• System evaluation: test valid and invalid API inputs, successful inference, endpoint/API errors, logging, IAM access, and dashboard rendering.

• Operational evaluation: record p50/p95 latency under stated test conditions and review CloudWatch logs/metrics.

• Project evaluation: compare delivered artifacts against the milestone checklist and document any scope reduction or deferred extension.

8. Suggested Improvements

Priority	Improvement	Reason / expected value	When
P0 — Must	Choose the exact dataset and prediction target early; document whether the task is binary failure classification, anomaly detection, or RUL.	Prevents mismatch between labels, algorithm, metrics and claims.	Before data preprocessing.
P0 — Must	Establish a leakage-safe split and baseline before tuning.	Makes reported test metrics more credible and reproducible.	During ML work.
P0 — Must	Create an end-to-end thin slice early: sample input → model prediction → API response → dashboard display.	Reduces integration risk and protects the final demo.	As soon as a baseline model exists; use mocks first if needed.
P0 — Must	Add data validation, controlled error messages, CloudWatch logging and least-privilege IAM checks.	Improves reliability and makes AWS failures diagnosable.	During API/cloud integration.
P1 — Should	Track experiment configuration, model version, threshold and dataset version.	Supports reproducibility and makes model comparisons fair.	During every model experiment.
P1 — Should	Use a threshold analysis focused on missed failures versus false alarms.	Makes evaluation more relevant to maintenance decisions than accuracy alone.	After candidate model evaluation.
P1 — Should	Add a cost checklist for endpoint startup/shutdown and verify no resources are left running.	Reduces avoidable AWS spending.	At every cloud session and before submission.
P2 — Later	Add real-time AWS IoT Core telemetry, automated alerts, QuickSight, multi-machine support or RUL.	Useful extensions but not required for the first working MVP.	Only after the core flow passes tests.

9. Final Grading Criteria

No official grading breakdown was included in the supplied guideline image or project proposal. The table below is a proposed internal self-assessment rubric only; it must not be represented as the instructor's official grade distribution. Replace it with the official rubric if the instructor provides one.

Assessment area	Proposed internal weight	Evidence expected
Documentation and project planning	25%	Clear proposal, scope, schedule/Gantt, milestones, roles, risk register, KPIs, literature review and requirements.
Implementation	35%	Reproducible data/ML pipeline, trained model, AWS architecture and deployed or demonstrably testable inference flow, dashboard.
Testing and evaluation	25%	Documented test cases, metrics, false-alarm analysis, API validation, error handling, IAM/security checks and monitoring evidence.
Presentation and demonstration	15%	Clear problem statement, architecture, live or recorded demo, results with limitations, team contribution and questions answered.
Total	100%	Proposed only; confirm official weights with the instructor.

9.1 Suggested Evidence Checklist

• Planning: approved scope, dated schedule, responsibilities, risks, KPIs and resource plan.

• Literature: cited sources, comparison of approaches, reason for dataset/model choice, recorded supervisor feedback where available.

• Requirements: stakeholder needs, user stories/use cases, FR/NFR IDs and acceptance criteria.

• Implementation: repository, setup instructions, data dictionary, preprocessing pipeline, model artifact and experiment log.

• Testing: test cases and results, confusion matrix, API response examples, error logs and monitoring evidence.

• Presentation: architecture diagram, clear end-to-end flow, known limitations, cost controls and next steps.

10. Requirements Gathering

10.1 Product Boundary and User Context

The MVP receives historical or simulated machine-condition data, validates and processes it, generates a failure-risk estimate or anomaly score, and presents results to maintenance users through a dashboard. It is not a safety-certified control system and must not automatically shut down machinery or replace qualified maintenance judgment.

10.2 Stakeholder Analysis

Stakeholder	Interest / need	Influence	Engagement approach
Maintenance engineer / operator	Understand machine status, see risk indicators, prioritize inspection and understand alert limitations.	High user influence	Review dashboard labels and example outputs; confirm what action guidance is useful.
System administrator / cloud owner	Secure, monitorable, cost-controlled AWS deployment; controlled access and recoverable errors.	High technical influence	Review IAM, logs, resource lifecycle and deployment steps.
Project team / developers	Clear requirements, manageable scope, reproducible development and integration.	High delivery influence	Weekly planning, Git/GitHub, code review and shared issue tracker.
Course instructor / supervisor	Evidence of learning, justified design decisions, documented process and clear presentation.	High assessment influence	Seek approval for scope, dataset choice and official rubric; record actual feedback.
Organization / equipment owner (future stakeholder)	Reliable information, low disruption, privacy/security and evidence of business value.	High in real deployment; advisory for MVP	Discuss real-world limitations; do not claim industrial benefit without field validation.

10.3 User Stories

ID	User story	Acceptance condition
US-01	As a maintenance engineer, I want to view each machine's latest sensor readings so that I can understand its current condition.	Dashboard shows the available sensor fields, units where known, timestamp and machine identifier.
US-02	As a maintenance engineer, I want to see a risk level or anomaly indicator so that I can prioritize inspection.	For valid input, the system displays the model output and clearly labels it as an estimate.
US-03	As a maintenance engineer, I want to see historical trends when data is available so that I can notice changes over time.	Dashboard plots available historical readings without inventing missing values.
US-04	As a user, I want invalid or incomplete input to be rejected clearly so that I know when the prediction cannot be trusted.	API returns a controlled validation error and does not produce a misleading prediction.
US-05	As a system administrator, I want errors and service health to be logged so that I can troubleshoot failures.	Relevant Lambda/API/endpoint errors and operational metrics are available through configured logs/monitoring.
US-06	As a project developer, I want repeatable model evaluation so that I can compare candidates fairly.	Dataset version, split, model configuration and evaluation metrics are recorded.

11. Use Cases

Use case	Actors	Preconditions	Main flow	Alternative / exception
UC-01: View machine health	Maintenance engineer	Dashboard is accessible; sample/historical data exists.	1. Open dashboard. 2. Select machine or sample. 3. View sensor readings, risk/anomaly output and suggested inspection cue.	If no data exists, show a clear empty state; do not show fabricated readings.
UC-02: Request a prediction	Dashboard user / API client	Inference model or test stub is available; request schema is known.	1. Submit validated sensor values. 2. API Gateway routes request to Lambda. 3. Lambda validates and invokes SageMaker endpoint. 4. Result is returned to dashboard.	Malformed input returns a validation error; endpoint timeout/service error is logged and returned as a controlled failure.
UC-03: Review abnormal behavior	Maintenance engineer	Prediction/anomaly output is available.	1. Review anomaly indicator or risk level. 2. Compare readings/trends. 3. Decide whether to inspect the machine.	The system presents a decision-support signal only; human staff make the maintenance decision.
UC-04: Diagnose a service failure	System administrator / developer	CloudWatch logging/monitoring is configured.	1. Check error/latency metrics. 2. Inspect logs. 3. Verify IAM and endpoint status. 4. Retry after corrective action.	If AWS is unavailable, use documented local test samples for demonstration and record the outage.
UC-05: Evaluate model version	ML developer	Versioned dataset and evaluation split exist.	1. Train candidate. 2. Evaluate on validation data. 3. Select threshold. 4. Run final test once. 5. Record metrics and limitations.	If labels are insufficient, report this limitation and evaluate anomaly detection with appropriate caveats.

11.1 Use Case Diagram (textual)

Actor	System interaction
Maintenance Engineer	→ View machine readings → View risk/anomaly result → Review trends → Decide whether to inspect
Dashboard / API Client	→ Submit sensor payload → Receive prediction or controlled error
System Administrator	→ Review CloudWatch logs/metrics → Diagnose endpoint/API/IAM problems
ML Developer	→ Prepare data → Train/evaluate model → Register selected model/version → Review metrics

11.2 Main Prediction Flow

1. Data source	2. Storage & ML	3. Inference	4. User
Public dataset or Python simulator	Amazon S3 → SageMaker preprocessing/training/evaluation	SageMaker endpoint ← Lambda ← API Gateway	Web dashboard displays readings, risk/anomaly output and action cue

Cross-cutting controls: AWS IAM limits permissions; Amazon CloudWatch captures operational logs and metrics. AWS IoT Core may be added for real-time telemetry, but direct S3 upload is acceptable for the MVP.

12. Functional Requirements

ID	Requirement	Priority	Verification
FR-01	The system shall accept machine-condition data containing the available fields required by the selected dataset, such as temperature, vibration, pressure, RPM, timestamp and machine ID.	Must	Submit a valid representative payload / load sample data.
FR-02	The system shall validate required fields, data types and acceptable value formats before inference.	Must	Test missing, malformed and wrong-type values.
FR-03	The preprocessing pipeline shall document and handle missing values, duplicates, outliers and categorical fields where relevant.	Must	Run data-quality tests and inspect preprocessing output.
FR-04	The training workflow shall split data in a way that avoids leakage; chronological or machine-group splitting shall be used when appropriate for the dataset.	Must	Review split code and data partitioning.
FR-05	The system shall train and compare at least one supervised baseline when failure labels support classification.	Must	Record model configuration and evaluation results.
FR-06	When labeled failure data is inadequate, the team shall evaluate anomaly detection as an alternative or complementary approach and state its limitations.	Should	Document data-label limitation and anomaly evaluation.
FR-07	The evaluation report shall include Precision, Recall, F1-score, confusion matrix and ROC-AUC when appropriate, plus false-alarm behavior.	Must	Inspect the final evaluation report.
FR-08	The system shall store raw/processed data and relevant model artifacts in Amazon S3, subject to configured permissions.	Must	Verify expected objects and access policy.
FR-09	The selected model shall be made available for inference through a SageMaker endpoint or a clearly documented test substitute if cloud deployment is blocked.	Must	Run a valid inference test and save evidence.
FR-10	The backend shall use AWS Lambda to validate/orchestrate prediction requests and call the deployed endpoint when available.	Must	Integration test and inspect logs.
FR-11	Amazon API Gateway shall expose a controlled API route for prediction requests.	Must	Send valid and invalid HTTP requests.
FR-12	The dashboard shall show available machine readings, prediction/risk or anomaly output, and a clear suggested inspection cue.	Must	Demonstrate dashboard with test data.
FR-13	The system shall log relevant API and inference errors and support operational monitoring through CloudWatch.	Must	Trigger a controlled error and verify log evidence.
FR-14	The system shall provide setup/deployment instructions and document required AWS services and configuration assumptions.	Must	Review documentation against a clean setup checklist.
FR-15	Real-time telemetry ingestion through AWS IoT Core may be added after the core MVP works.	Could	Demonstrate MQTT/data flow only if implemented.

13. Non-functional Requirements

ID	Quality attribute	Requirement / target	Verification
NFR-01	Performance	Proposed API target: p95 response time ≤ 3 seconds under a documented student-demo test workload; this is a target, not a current result.	Run repeatable requests; report sample count, environment and p50/p95.
NFR-02	Model quality	Proposed initial goals are Recall ≥ 0.80 and F1-score ≥ 0.75 when supported by the chosen dataset; actual results must be reported without manipulation.	Evaluate on held-out data and record positive-class support.
NFR-03	Security	Use IAM least-privilege roles; do not commit secrets; restrict S3 and endpoint access to required identities.	Review IAM policies, repository and bucket access.
NFR-04	Reliability	Invalid requests and cloud failures shall result in controlled error responses rather than unhandled crashes or fabricated predictions.	Test malformed payloads, endpoint timeout/error and missing data.
NFR-05	Observability	Log API/inference errors and relevant request outcomes; use CloudWatch metrics/logs for operational troubleshooting.	Inspect CloudWatch evidence and log configuration.
NFR-06	Usability	Dashboard terminology shall be understandable to a non-ML maintenance user; risk levels and anomaly scores must be labelled clearly.	Walk through user stories and gather reviewer feedback.
NFR-07	Maintainability	Separate data preprocessing, model training, API logic and dashboard code; document dependencies and setup steps.	Repository review and setup walkthrough.
NFR-08	Reproducibility	Record dataset source/version, split method, model parameters, threshold and evaluation metrics.	Re-run documented evaluation or inspect experiment log.
NFR-09	Cost efficiency	Avoid unnecessary always-on compute; stop unused SageMaker endpoints and review AWS usage after sessions.	Resource checklist and billing/usage review.
NFR-10	Scalability	Architecture should permit later ingestion of real telemetry and additional machines, but MVP need not prove production-scale capacity.	Document extension points; load testing is optional.
NFR-11	Data integrity	Preserve raw data separately from processed data and document any filtering, imputation, encoding or resampling.	Inspect S3 structure and preprocessing documentation.
NFR-12	Safety / human oversight	Predictions are advisory. The application shall not automatically control or shut down equipment; maintenance decisions remain with qualified personnel.	Review UI text and project scope.

13.1 Data and API Contract (initial)

Element	Expected behavior
Input	JSON payload with the required features for the selected dataset; exact field names and units must be documented in the implementation.
Successful response	A prediction label or risk score, optional anomaly score, model/version identifier where practical, and a concise interpretation.
Invalid request	A clear 4xx-style validation response; no inference on malformed input.
Service failure	A controlled service error, appropriate log entry, and no fabricated prediction. Retry behavior should be limited and documented.
Limitations	Output is only as reliable as the data, model validation and threshold selection; it is not a guarantee of failure or safety.

14. Acceptance Criteria & Requirements Traceability

The MVP is considered ready for demonstration when the Must requirements below have evidence. A requirement may be marked complete only after the team records the relevant test, screenshot, log, or document. A failed test should be recorded with an issue and corrective action rather than hidden.

Requirement group	Acceptance criteria	Evidence
Data (FR-01 to FR-04; NFR-08, NFR-11)	Dataset source and target are documented; data checks run; train/validation/test split is reproducible and leakage-aware.	Data dictionary, validation report, split code.
ML (FR-05 to FR-07; NFR-02)	At least one suitable baseline is evaluated; metrics and false alarms are reported; limitations are stated.	Experiment log, confusion matrix, metrics table.
Cloud inference (FR-08 to FR-11; NFR-03 to NFR-05)	S3/SageMaker/API flow works in the available environment or fallback is explicitly documented; invalid inputs and failures are handled.	API tests, IAM review, CloudWatch logs, deployment notes.
Dashboard (FR-12; NFR-06, NFR-12)	A user can see readings and output; labels are understandable; human oversight is explicit.	Dashboard screenshot or demo checklist.
Documentation (FR-14; NFR-07, NFR-09)	A teammate can follow setup instructions; AWS resources are listed and idle resources are shut down after use.	README, resource checklist, final report.

14.1 Final Submission Checklist

- ■ Project proposal, scope and objectives are consistent with the original proposal.
- ■ Gantt chart includes dates, workstreams, milestones and deliverables.
- ■ Every team member has a primary role and named outputs.
- ■ Risk register includes data quality, rare failures/class imbalance, time-series leakage, AWS cost, overengineering, connectivity/service failures, and IAM permissions.
- ■ KPIs have measurement methods and clearly marked proposed targets; actual results are not fabricated.
- ■ Literature review cites relevant sources and explains how they inform design choices.
- ■ Supervisor feedback is recorded only after received; official grading rubric is used if provided.
- ■ Stakeholders, user stories, use cases, FRs, NFRs and acceptance criteria are included.
- ■ All AWS resources and credentials are handled safely; no secrets are included in the submitted PDF/repository.
- ■ Final PDF and presentation are proofread, consistent, and include evidence where implementation is claimed.

Sign-off (to be completed)

Role	Name	Signature / date
Team representative	Abdelrahman Ahmed	_____
Supervisor / instructor	To be completed	_____

15. References

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[10] IEEE. (2021). Challenges of machine learning-based RUL prognosis: A review on NASA's C-MAPSS data set. In *2021 26th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*. https://doi.org/10.1109/ETFA45728.2021.9613682

Reference use note: these sources support the project framing and design choices. Dataset suitability, target labels and final metrics must still be verified against the exact dataset version selected by the team.

Appendix A — Original AWS Architecture Baseline

Layer	Technology	Role in the proposed MVP
Data source	Public dataset / Python simulator	Provides historical or simulated telemetry such as temperature, vibration, pressure, RPM, operating hours and timestamps.
Storage	Amazon S3	Stores raw/processed data, model artifacts and prediction outputs.
ML	Amazon SageMaker	Supports training, evaluation, deployment and inference.
Backend	AWS Lambda	Validates requests and orchestrates calls to the inference endpoint.
API	Amazon API Gateway	Provides a controlled prediction interface.
Monitoring	Amazon CloudWatch	Collects logs, errors and operational metrics.
Security	AWS IAM	Enforces least-privilege access between components.
Optional ingestion	AWS IoT Core	Receives real-time telemetry if the team implements the extension.

Appendix B — Important Definitions

Term	Meaning in this document
MVP	Minimum Viable Product: the smallest end-to-end version that demonstrates data → model → API → dashboard.
Recall	Share of actual failure cases detected by the model; important when missed failures are costly.
Precision	Share of predicted failures that are actual failures; relevant to false alarms.
F1-score	Harmonic mean of Precision and Recall.
ROC-AUC	A threshold-independent ranking metric for binary classification when appropriate labels and scores are available.
p95 latency	Response time at or below which 95% of measured requests complete under the stated test conditions.
Time-series leakage	Use of information during training/evaluation that would not legitimately be available at prediction time.